GOOGLE QAI

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Team



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Problem

- ➔ Prepare the Dicke state D(N, n): an equal superposition of every N-bit string with exactly n ones.
- ➔ Allowed gates: CZ and single-qubit PhasedXZ (plus optional reset). CZ only between neighboring qubits on the grid.
- ➔ Error rates differ by qubit and coupler. In one sample calibration, the worst coupler is 20× worse than the best.
- ➔ Scored on Willow hardware from 10,000 shots in the Z basis and 10,000 in the X basis.

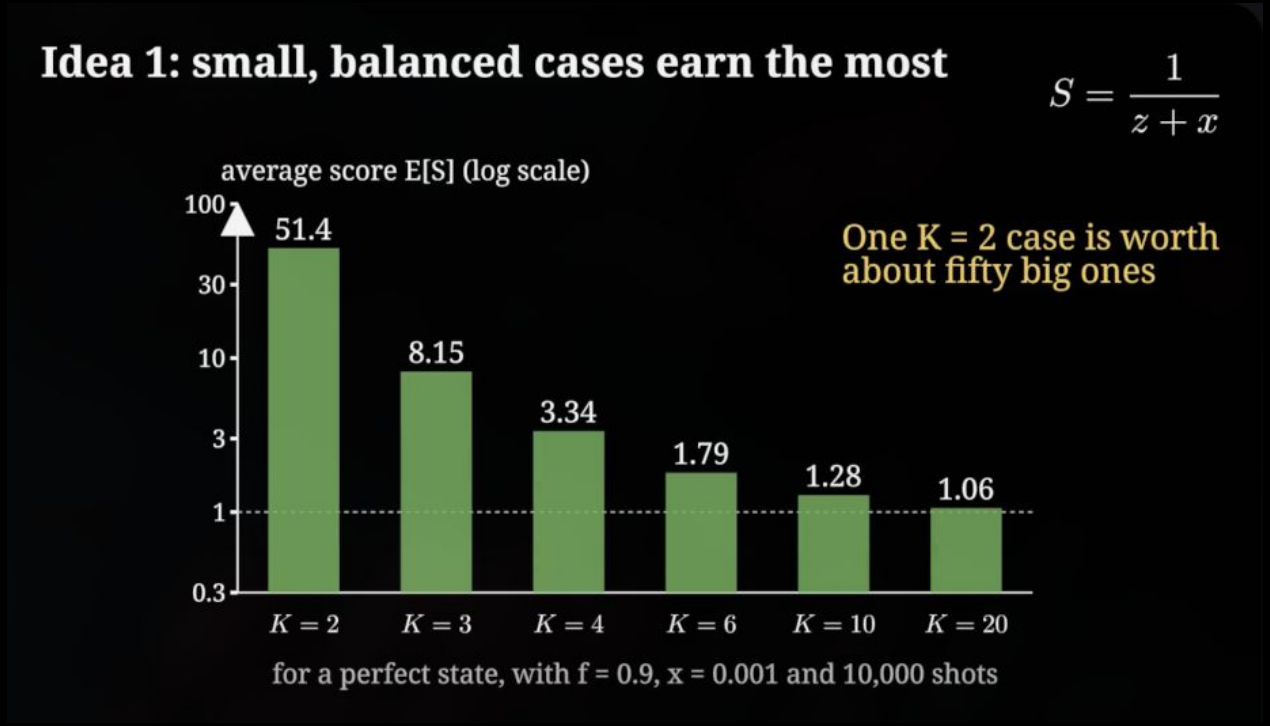
Evaluation harness

- ➔ Noise model built from the calibration data: depolarizing error after each gate, asymmetric readout error.
- ➔ The judges' scoring function, copied verbatim, on 10,000 simulated shots per basis.
- ➔ Score broken into leakage, uniformity, and X-basis (phase) error, and compared with the score of a perfect state.
- ➔ Tuned on calibrations 0–2, checked on held-out calibrations 3–5. Fixed seeds; every result saved to CSV.

First results

- ➔ Textbook circuit (Bärtschi–Eidenbenz) for 8 qubits, 4 excitations: 24% of a perfect state's score.
- ➔ It assumes any two qubits can interact. Routing it onto the grid adds SWAPs (3 CZ each): 162 CZ in total.
- ➔ At about 0.25% error per CZ, roughly a third of the fidelity is lost to CZ errors alone.
- ➔ So the main goal became fewer CZs, placed on the best qubits.

Create a Cirq circuit to produce a Dicke state



Lower k-values are more important to optimize for

PICKING CIRCUITS

n = 0 or n = N

→ Product state, no entanglement

n = 1

→ Splitting tree, depth ~ log N

N ≤ 8

→ Computer-searched exact circuit

Larger N

→ Two-row split, rows built in parallel

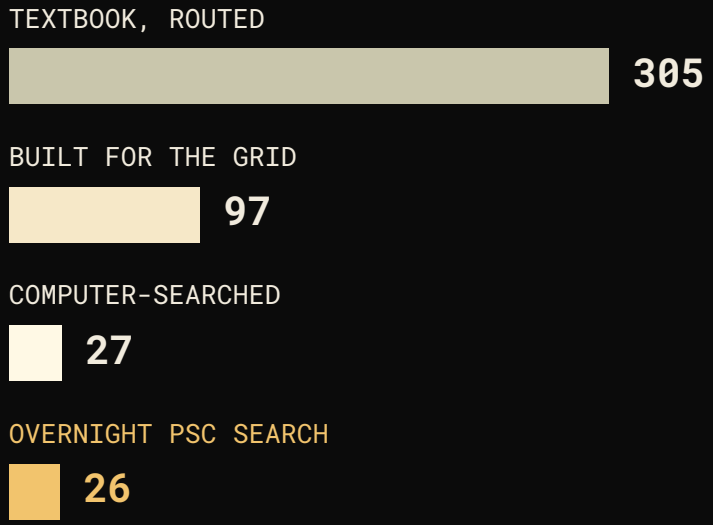
Idea 2: spread a W state along a tree, not a line

Reduce moments by using trees

Reducing gate count

Same exact state, 12x fewer two-qubit gates.

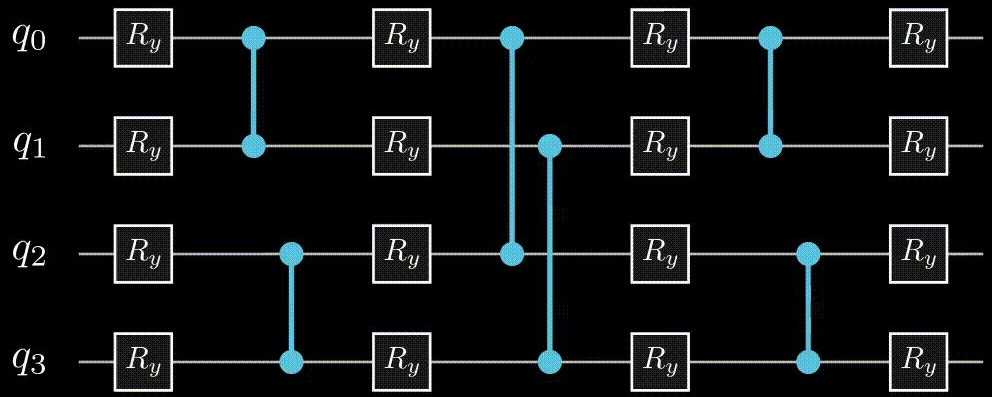
CZ gates for D(8,4): 8 qubits, 4 excitations



Every CZ carries every kind of error, so removing one removes all of them.

Idea 3: let a computer search for exact circuits

goal: $|D_2^4\rangle$ building blocks: R_y rotations and CZs between neighbours

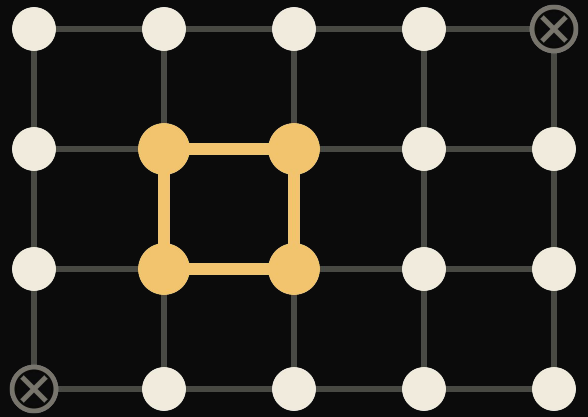


CZ: 6

Remove one CZ and re-fit.
Still exact? Leave it out.
Otherwise, put it back.

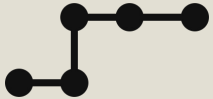
Cost Function for Placements

- → Remove broken qubits
- → Cost = gate errors + readout errors
- → Penalty for mismatched readout
- → Pick least cost placement



Yellow: Min cost

SEARCHING THE CHIP



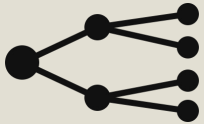
LINES

Grow paths one qubit at a time, keep the 256 cheapest



SHAPES

Try every spot, in all 8 rotations and mirrors



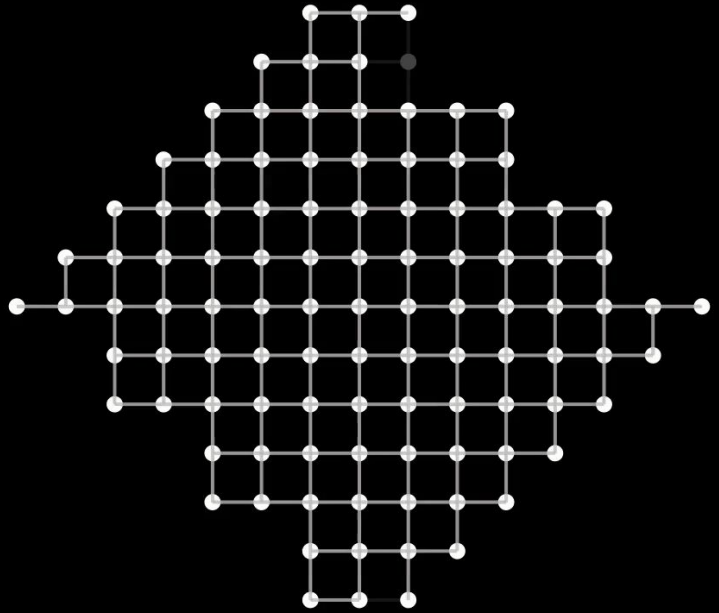
TREES

Grow outward from the best starting qubit

Swapping qubits doesn’t change a Dicke state, so any spot that fits is a valid answer.

Tested with up to 5% of qubits broken: every case up to N = 30 builds a valid circuit

Idea 4: put the circuit on the best qubits



Demo

Questions?